# PHAN VAN BANG

Game Developer · Unity, C#, Online Games

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## SUMMARY

Gameplay and online-game programmer (Unity / C#) with a finished 3D action-RPG prototype, a 2D platformer and an OpenGL racing game. Built the networking layer of **Eternal Odyssey** on Firebase Realtime Database, NavMesh enemy AI, combat, inventory and quests, and connected the game to an AI lore assistant that streams answers into the in-game UI. Also writes the backend and mobile clients that games need (Node.js, React Native, Firebase).

## TECHNICAL SKILLS

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|---|---|
| **Game** | Unity 3D and 2D, C#, Animator state machines, NavMesh AI, coroutines, TextMesh Pro, Cinemachine, VFX and particle integration, custom Editor tools |
| **Online** | Firebase Auth and Realtime Database, state interpolation for remote players, synced world objects and chests, REST and streaming HTTP from Unity |
| **Low level** | C++, OpenGL / GLUT, C, operating-systems coursework |
| **Mobile & backend** | React Native, Android (Java), Node.js / Express, MongoDB, Firebase |
| **AI in games** | RAG lore assistant (FastAPI, Qdrant, GPT-4o) with SSE streaming into Unity |

## SELECTED PROJECTS

### Eternal Odyssey – Unity 3D action RPG (team of two)

Unity · C# · NavMesh · Firebase Realtime DB · TextMesh Pro

- **Scope:** 178 first-party C# scripts in 12 domain folders: characters, controllers, models (28 data classes and 19 enums), inventory, quests, realtime, systems and UI.
- **Online play:** custom Firebase sync instead of a networking SDK: login and registration, player spawn and room logic, Lerp-interpolated remote players with animation-state sync, shared world objects and chest contents.
- **Combat and enemies:** NavMesh patrol / chase / attack with sight and attack ranges and Animator state behaviours; enemy combo counter that picks from 7 skills with min/max damage and critical chance; player base attack, shield, projectile and area skills; a buff and debuff API on a rich attribute model (damage, health, speed, crit).
- **Systems:** drag-and-drop inventory with item actions, ground items and chests; quest manager, quest world and quest UI; custom Editor inspectors for quest and weapon lists.
- **AI integration:** in-game assistant posts player questions to a RAG API and prints the streamed answer character by character, with spoiler-aware persona prompts.
- **Design:** ER diagram and use-case model (StarUML) plus a game-design document: 5 elemental classes, 3-tier weapons with proc effects and EXP rewards scaled by enemy tier.

### No Than – Unity 2D platformer (mobile and desktop)

Unity 2D · C# · Joystick · Cinemachine

- 39 gameplay scripts: physics movement with double jump and timed dash with speed boost and VFX, mobile joystick plus keyboard input, two player skills, enemies with contact damage and aimed projectiles, hearts, traps, rewards, NPC dialogue, boss-kill flow and scene transitions.

### Racing Game – C++ and OpenGL

github.com/pvbang/cpp-racing-game

- Endless scrolling-road racer in GLUT: randomised non-overlapping obstacle cars, collision test, score and speed levels, cloud layer and HUD. Includes a demo video.

### The Escape – React Native game with Node.js API

- Mobile game client in React Native with its own Node.js backend (game, API).

### CV World – browser voxel game (Three.js)

pvbang.github.io/cv/game

- A Minecraft-style 3D CV you can walk through: terrain and buildings, mining and block placing, mobs and combat, collectible shards, NPC dialogue, fast travel, pause menu and a quick auto tour; playable on desktop and mobile with no install.

## EDUCATION & LANGUAGES

**B.Eng. in Information Technology** · Vietnam-Korea University of ICT (VKU), Da Nang — 2020 – 2025

- Languages: Vietnamese (native), English (professional working proficiency: technical reading and writing).